

Exact Noisy Support Cutoff Fails in Oversmoothing ℓ^1 Regularization

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Status. Candidate counterexample, likely valid. The result disproves the exact support-localization route proposed in the source; it does not disprove convergence rates or quantitative tail bounds.

1 Source question and its precise reading

Gerth and Hofmann consider oversmoothing ℓ^1 regularization when the exact solution belongs to $\ell^2 \setminus \ell^1$. In their final discussion they describe an incomplete argument based on truncations $P_n x^\dagger$ and state that the open problem is to show that “the support of the approximate solutions is not larger than this $n(\delta)$ ” [1, printed pp. 18–19]. The source passage is reproduced below.

$$\varphi(t) = 2 \inf_{n \in \mathbb{N}} \left(\sum_{k=n+1}^{\infty} |x_k^\dagger| + K t \right) = K t.$$

This, however, yields by Proposition 3.7 the rate (11) and completes the proof of the proposition. $\square$

Property 4.2 enables us to show convergence rates for ℓ^1 -regularization for ill-posed and well-posed problems with sparse and non-sparse solutions. It has been shown in 17 that the rate function φ in (14) does in general not saturate. Even more, the rate is always obtainable either with an a priori or an a posteriori choice of the regularization parameter. This stands in sharp contrast to classical Tikhonov regularization, i.e., (3) with $p = 2$ and $\|x\|_{\ell^2}$ as penalty, which is known to admit convergence rates up to a maximum of $\delta^{2/3}$ for a suitable a priori choice of the regularization parameter and only a rate of $\delta^{1/2}$ under the discrepancy principle. Since the smoothness of the solution is typically unknown, this makes ℓ^1 -regularization more attractive from the viewpoint of regularization theory than its ℓ^2 counterpart. One simply uses the discrepancy principle and no longer has to care about the smoothness of the solution. However, one may run into trouble when the solution does not belong to ℓ^1 but only to $\ell^2 \setminus \ell^1$ such that $\|x^\dagger\|_{\ell^1} = \infty$. In such a case we call the regularization method (3) *oversmoothing*.

There are promising results showing that even in the situation of oversmoothing, ℓ^1 -regularization may lead to convergence rates in a weaker norm. Again, an a priori choice or the discrepancy principle for the choice of α would lead to the optimal rates. Preliminary results have been shown in the preprint 15. There, a strategy is shown to derive convergence rates in the ℓ^2 -norm for ℓ^1 -regularization for every $x^\dagger \in \ell^2$. The proof of a theorem analogously to Proposition 3.7 unfortunately was incomplete. It revolves around approximating $x^\dagger$ with $P_n x^\dagger$ and letting $n = n(\delta) \rightarrow \infty$ as $\delta \rightarrow 0$ with a specific choice of $n = n(\delta)$. The open problem was to show that the support of the approximate solutions is not larger than this $n(\delta)$. It appears that such a statement is possible by using item (c) in Property 4.2 and the necessary optimality condition for a minimizer of (3), where the latter provides us with the norm $\|\eta\|_{Y^*}$ in Property 1

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corresponding to a $\xi = A^* \eta \in \partial \|x_\alpha^\delta\|_{\ell^1}$. Since we are not able to use a variational source condition when $\|x^\dagger\|_{\ell^1} = \infty$ we need to use a different approach to show convergence rates. To this end we seem to have a chance to adapt the strategy of 17 based on elementary steps. Consequently, we hope to complete the detailed proof in an upcoming paper 16.

The truncation argument needs the exact initial-segment localization

$$\text{supp}(x_\alpha^\delta) \subseteq \{1, \dots, n(\delta)\}, \quad \text{equivalently} \quad (I - P_{n(\delta)})x_\alpha^\delta = 0. \quad (1)$$

We show that (1) is false under the stated deterministic noise model, already in the polynomial diagonal example used by the announced follow-up paper [2].

2 Idea of the proof

For a diagonal forward operator the ℓ^1 -regularized minimizer is obtained by coordinatewise soft thresholding. The follow-up parameter scaling makes the threshold in data coordinate $n + 1$ smaller than the allowed noise amplitude δ : their ratio is at most a fixed constant divided by $\sqrt{n}$. Placing the whole admissible noise vector in coordinate $n(\delta) + 1$ therefore forces that coordinate to survive thresholding. Exact support cutoff is thus too brittle for worst-case deterministic noise; a tail-norm estimate is the appropriate replacement.

3 Counterexample

Fix $\beta > 0$ and $\eta \in (1/2, 1]$. On ℓ^2 let

$$Ae_k = \sigma_k e_k, \quad \sigma_k = k^{-\beta}, \quad x_k^\dagger = k^{-\eta}, \quad y = Ax^\dagger.$$

Then A is compact and injective, while $x^\dagger \in \ell^2 \setminus \ell^1$. Put

$$T_m = \|(I - P_m)x^\dagger\|_2, \quad \varphi(\delta) = \min_{m \geq 1} \left(\frac{\delta}{\sigma_m} + T_m \right),$$

and choose any minimizing index $n(\delta)$. This is the prototype rate and truncation index of [2, Section 4]. For a fixed constant $c_\alpha > 0$, take the same a priori scaling

$$\alpha(\delta) = \frac{c_\alpha \delta^2}{\sqrt{n(\delta)} \varphi(\delta)}. \quad (2)$$

For data y^δ , let x_α^δ minimize

$$\frac{1}{2} \|Ax - y^\delta\|_2^2 + \alpha \|x\|_1 \quad \text{over } x \in \ell^2. \quad (3)$$

Theorem 1. *For every fixed $c_\alpha > 0$ and every sufficiently small $\delta > 0$, there is admissible data y^δ with $\|y^\delta - y\|_2 = \delta$ for which*

$$n(\delta) + 1 \in \text{supp}(x_\alpha^\delta).$$

Consequently, the exact support cutoff (1) fails even in this diagonal oversmoothing model.

Proof. First, $n(\delta) \rightarrow \infty$ as $\delta \downarrow 0$. Indeed, for each fixed M , $T_M > 0$. Choose $m > M$ with $T_m < T_M/2$ and then choose δ so small that $\delta/\sigma_m < T_M/2$. The objective at m is then smaller than its limit T_M at every index at most M , after decreasing δ once more if needed. Thus no bounded sequence of indices can minimize for arbitrarily small δ .

Write $n = n(\delta)$. Since $\varphi(\delta) \geq \delta/\sigma_n$, equation (2) gives

$$\frac{\alpha}{\delta \sigma_{n+1}} \leq \frac{c_\alpha}{\sqrt{n}} \frac{\sigma_n}{\sigma_{n+1}} = \frac{c_\alpha}{\sqrt{n}} \left(1 + \frac{1}{n}\right)^\beta \leq \frac{c_\alpha 2^\beta}{\sqrt{n}}. \quad (4)$$

The right-hand side tends to zero. Hence $\alpha < \delta\sigma_{n+1}$ for all sufficiently small δ .

Now set

$$y^\delta = y + \delta e_{n+1}.$$

This data vector has exactly the allowed noise norm. Since (3) decouples in the diagonal basis, its unique minimizer is given by soft thresholding:

$$(x_\alpha^\delta)_k = S_{\alpha/\sigma_k^2}\left(\frac{y_k^\delta}{\sigma_k}\right), \quad S_\lambda(t) = \text{sgn}(t)(|t| - \lambda)_+.$$

At $k = n + 1$ all terms are positive, and therefore

$$(x_\alpha^\delta)_{n+1} = x_{n+1}^\dagger + \frac{\delta}{\sigma_{n+1}} - \frac{\alpha}{\sigma_{n+1}^2} > 0,$$

where the last inequality follows already from $\delta\sigma_{n+1} > \alpha$. Thus the minimizer has nonzero support strictly beyond $n(\delta)$, proving the claim. $\square$

4 Relation to the follow-up literature

The announced follow-up [2] restricts the general program to compact diagonal operators. Its noisy-data argument does not assert exact vanishing beyond $n(\delta)$; instead Lemma 4.4 estimates $\|(I - P_{n(\delta)})x_\alpha^\delta\|_2$. Theorem 1 explains why this weakening is necessary under worst-case deterministic noise. The counterexample does not challenge that quantitative estimate or the resulting convergence-rate theorem.

5 Verification and novelty bounds

The script `code/verify_counterexample.py` evaluates the exact soft-threshold formula for $\beta = 1$, $\eta = 3/4$, $c_\alpha = 10$, and five noise levels from 10^{-4} to 10^{-6} . In all cases the coefficient at $n(\delta) + 1$ is positive. This computation is only a sanity check; the proof above is analytic.

A bounded search through 27 August 2026 used arXiv, the exact source and follow-up titles, the authors’ publication page, the exact source phrase, and combinations of “support”, “noise”, “counterexample”, “correction”, “erratum”, and n_{inf} . It found the 2019 follow-up and later oversmoothing work, but no explicit record of this counterexample or an erratum addressing the exact cutoff. The novelty claim is therefore only that the argument appears unrecorded within this bounded search.

6 Limitations and review recommendation

The result addresses support *localization* in the first $n(\delta)$ coordinates, the interpretation required to make the P_n tail vanish. It does not rule out a statement only about the cardinality of the support. A reviewer should first confirm this interpretation of the source wording; after that, the only substantive checks are the soft-threshold formula and estimate (4).

References

- [1] D. Gerth and B. Hofmann, *On ℓ^1 -regularization under continuity of the forward operator in weaker topologies*, in New Trends in Parameter Identification for Mathematical Models, Birkhäuser, 2018, pp. 67–88; arXiv:1711.08642.

- [2] D. Gerth and B. Hofmann, *Oversmoothing regularization with ℓ^1 -penalty term*, AIMS Mathematics 4 (2019), 1223–1247, doi:10.3934/math.2019.4.1223.